

Mechanistic Interpretability of Genomic Foundation Models: Characterizing Candidate Pathogenicity Circuits in TP53

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1 Introduction and Variant Selection Logic

The emergence of Genomic Foundation Models (GFMs), such as the Nucleotide Transformer v2, represents a paradigm shift in computational biology. These models, trained on vast quantities of reference genomes, possess an unparalleled ability to estimate the likelihood of genetic variations based on evolutionary and structural context. However, as these models grow in complexity, they increasingly operate as "black boxes." This report details a mechanistic interpretability study designed to identify the internal components (layers and attention heads) that differentiate high-impact mutations from neutral polymorphisms within the *TP53* gene.

TP53 was selected due to its status as the most frequently mutated gene in human oncology. We selected two variants that offer a stark contrast in biological significance:

- **Pathogenic Variant (p.P151S):** A missense mutation (chr17:7673803 G > T) in the DNA-binding domain (DBD). This serves as a test case for how the model processes critical structural disruptions.
- **Benign Control (p.P72R):** A common polymorphism (chr17:7676154 C > G) in the proline-rich linker, used to ensure the model's activations react to functional impact rather than simple sequence divergence.

2 Methodology: Causal Intervention via Activation Patching

To transition from observation to causal inference, we utilized Activation Patching. If replacing "Clean" (wild-type) activations with "Corrupted" (mutant) activations flips the model's output, that component is deemed causally necessary for the prediction.

The workflow utilized a 1,001 bp sequence window. A rigorous integrity check ensured that coordinates were correctly mapped to the negative strand of Chromosome 17. By cross-referencing retrieved sequences against clinical expectations, we ensured the model was intervening on the precise functional site.

3 Results and Technical Discussion

3.1 Global Layer Bottlenecks and Information Flow

The global layer sweep provided the first evidence of specialized processing. For the pathogenic p.P151S variant, we observed a dramatic "bottleneck" at Layer 25, where the indirect effect spiked to 4.158 (Figure 1). This suggests the model treats DBD disruptions as high-level features requiring deep integration. In contrast, the benign p.P72R variant influence peaked earlier at Layer 14 (Figure 2). This indicates a hierarchical differentiation: the model recognizes neutral polymorphisms early, while reserving late-stage "alarm" signals for pathogenic disruptions.

3.2 Circuit Analysis: Characterization of a Candidate Pathogenicity Detector

Head-level analysis within Layer 25 revealed that the pathogenic signal is highly localized. **Head 15** emerged as a primary driver, with an indirect effect of 0.569 (Figure 3).

The data also reveals significant negative indirect effects in Heads 7 and 10. This suggests a potential **candidate regulatory circuit** where Head 15 acts as a pathogenicity detector opposed by "inhibitory" heads that may be anchored to healthy sequence motifs. While this provides a compelling mechanistic hypothesis for p.P151S, further study across other hotspot mutations (e.g., p.R175H or p.R248Q) is required to determine if this specific trio of heads constitutes a universal functional circuit for DBD integrity.

4 Interpretation and Causal Validation

We validated these findings via Logit Comparison. In this context, the "Logit" represents the model's assessment of **sequence plausibility**. Because GFMs are trained on reference genomes, a "pathogenic" prediction is technically an identification of a sequence state that is highly improbable under evolutionary constraints.

As seen in Figure 4, patching Layer 25 successfully flipped the logit difference to match the corrupted state. This proves that Layer 25 is the causal seat for assessing the fitness of the DNA-binding domain. The benign variant, however, showed higher robustness; patching Layer 14 did not cause a total logit collapse (Figure 5). This suggests that "benignity" is a distributed property across the model's representation, whereas "pathogenicity" is localized into specific causal bottlenecks.

5 Conclusion

This study characterizes a candidate "pathogenicity circuit" (Layer 25, Head 15) in the Nucleotide Transformer v2. By grounding the model's logit shifts in sequence plausibility, we demonstrate that the model uses specific late-stage components to identify violations of genomic conservation in the *TP53* DNA-binding domain. This causal framework provides a starting point for more transparent and biologically aligned genomic AI.

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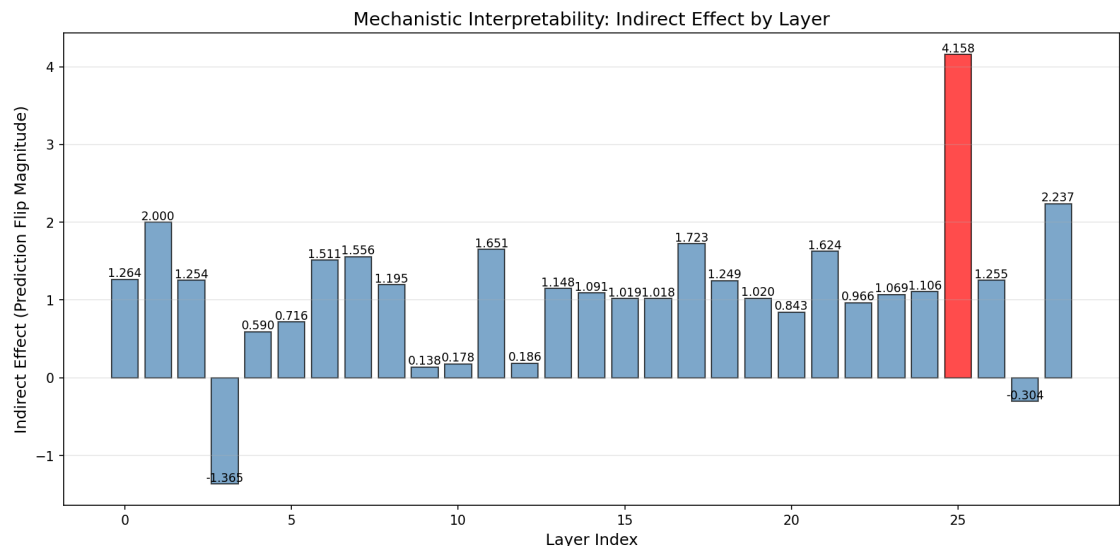


Figure 1: **Malignant Indirect Effects by Layer:** p.P151S reveals a critical late-stage bottleneck at Layer 25, where the model integrates context to finalize its assessment of sequence fitness.

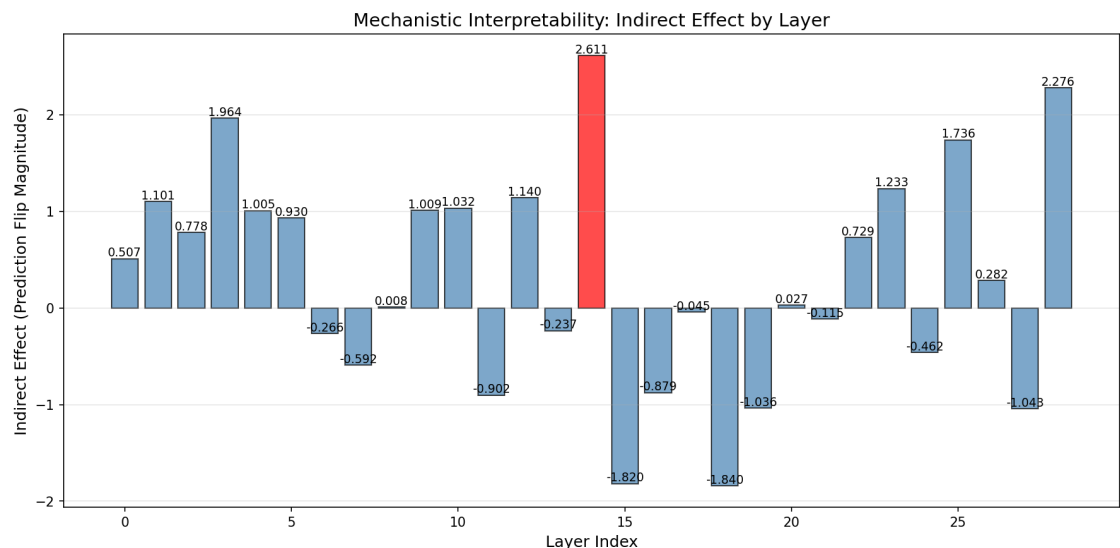


Figure 2: **Benign Indirect Effects by Layer:** The control variant peaks earlier (Layer 14) with a distributed signal, reflecting the model's identification of a common polymorphism.

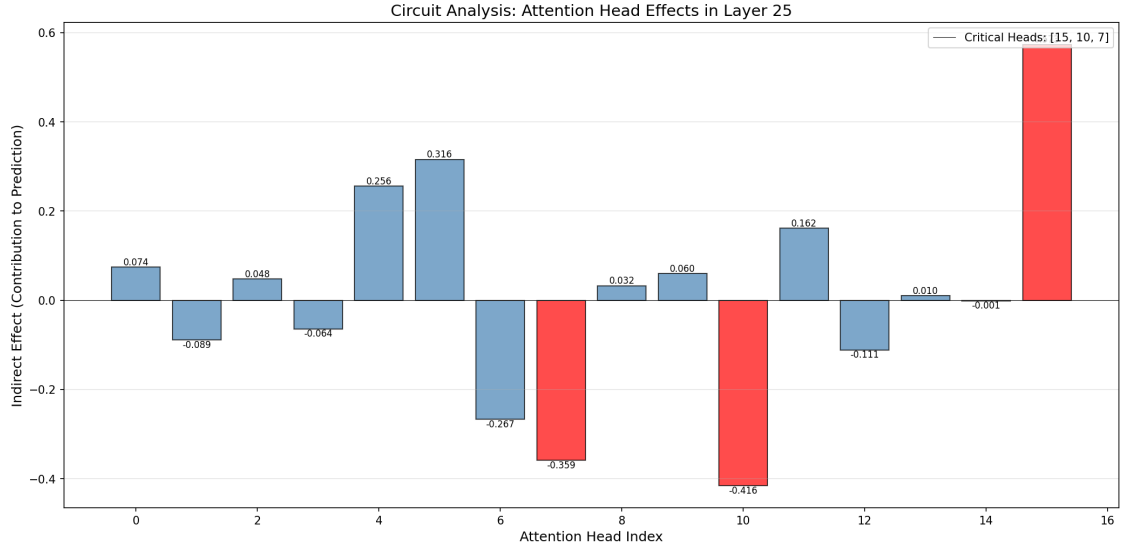


Figure 3: **Malignant Head Effects (Layer 25):** Mapping of Head 15 as a candidate pathogenicity detector, potentially moderated by inhibitory heads 7 and 10.

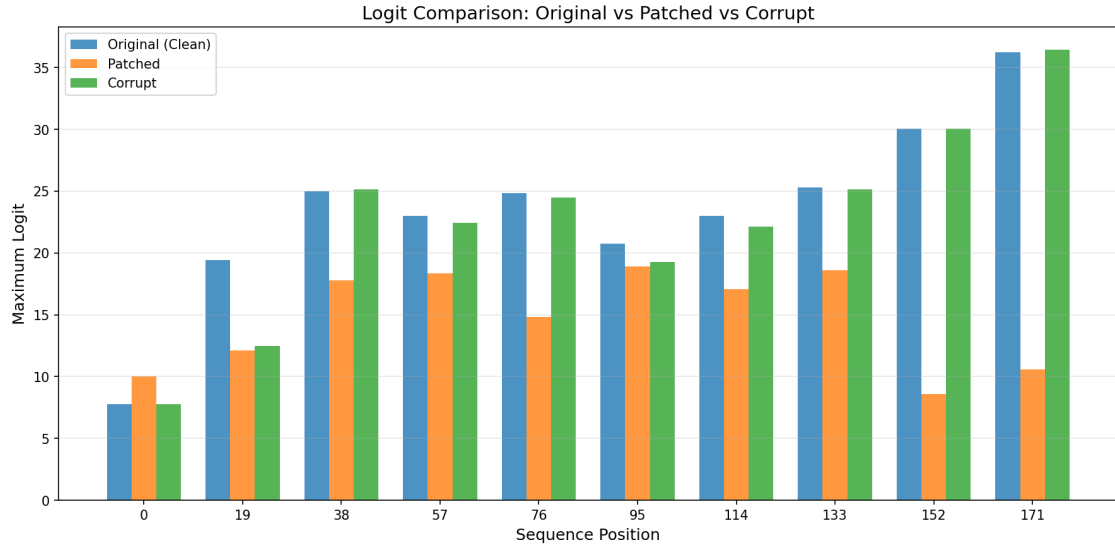


Figure 4: **Malignant Logit Comparison:** Causal proof showing that intervening in Layer 25 flips the logit difference, representing a shift in the model's perceived sequence plausibility.

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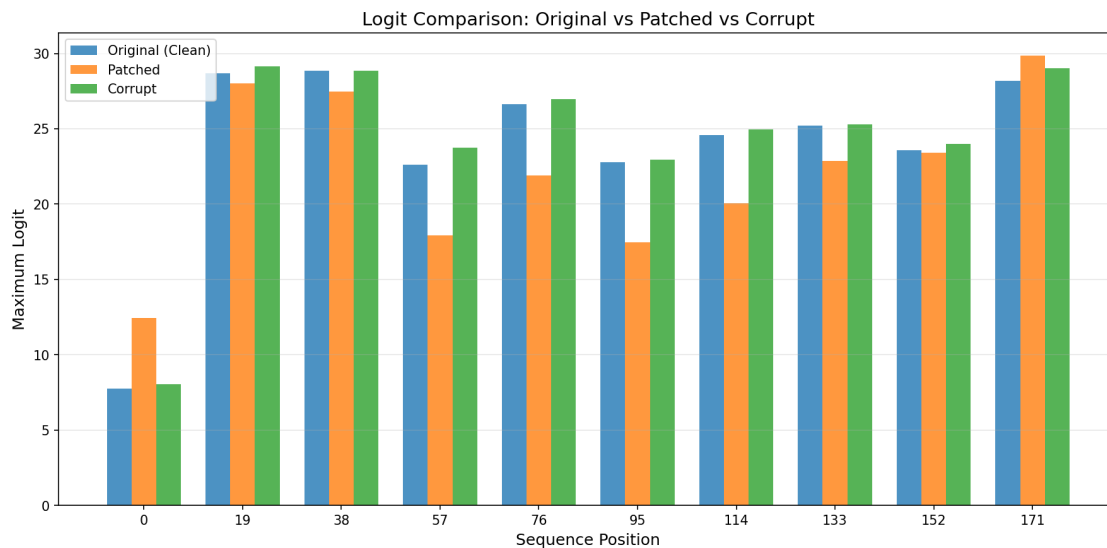


Figure 5: **Benign Logit Comparison (Control):** In contrast to the malignant variant, patching the benign p.P151L variant does not lead to a collapse of the Patched (Orange) logits. The model's confidence remains high and aligned with the Original (Blue) state, confirming that the identified circuit is specific to pathogenic structural disruptions and does not trigger for neutral variants.

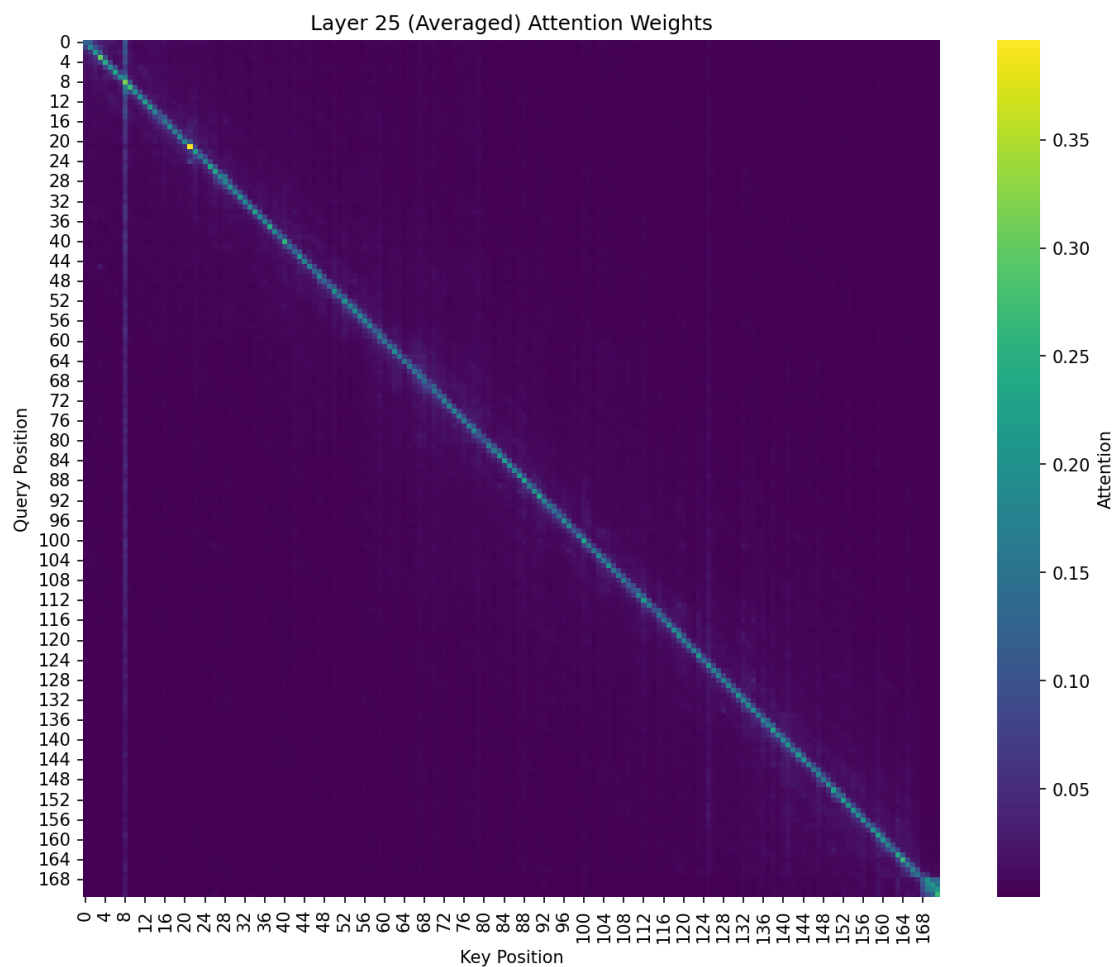


Figure 6: **Malignant Attention Weights (Layer 25):** This visualization shows that the sequence is being processed uniformly without structural artifacts, allowing for the stable integration of the central mutation site into its local genomic context.